

MUROTOMISAKI, Japan

WMO: 478990

Lat: 33.2522N

Lon: 134.1772E

Elev: 187

StdP: 99.10

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	1.2	2.4	-9.7	1.7	2.7	-8.1	1.9	3.6	22.4	8.5	20.2	7.9	9.3	310	0.668

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	4.0	29.7	26.1	29.0	25.8	28.3	25.4	26.9	28.7	26.4	28.1	26.0	27.5	5.0	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
26.4	22.4	27.9	26.0	21.8	27.4	25.5	21.2	27.0	85.7	28.8	83.7	28.1	81.8	27.6	28.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
20.2	17.2	14.6	-1.0	31.0	1.4	0.8	-1.9	31.6	-2.7	32.0	-3.5	32.4	-4.5	33.0		
			-4.0	27.7	1.2	0.6	-4.9	28.1	-5.6	28.4	-6.3	28.8	-7.2	29.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	17.1	7.6	8.3	11.0	15.2	19.1	21.6	25.4	26.6	24.2	20.0	15.1	10.2
	DBStd	6.91	2.71	3.06	2.99	2.55	1.74	1.79	1.63	1.30	1.93	2.28	2.92	3.08
	HDD10.0	212	83	66	24	1	0	0	0	0	0	1	37	
	HDD18.3	1317	332	282	227	97	13	1	0	0	10	102	254	
	CDD10.0	2793	9	17	56	157	281	349	477	516	427	309	154	42
	CDD18.3	856	0	0	0	2	36	100	218	257	177	61	5	0
	CDH23.3	4926	0	0	0	0	9	159	1492	2275	924	66	1	0
	CDH26.7	692	0	0	0	0	0	3	191	411	86	1	0	0
Wind	WSAvg	7.2	7.8	7.5	7.7	7.3	6.9	6.9	6.7	6.4	7.1	7.4	7.1	7.6
Precipitation	PrecAvg	2597	92	122	176	195	254	350	294	268	360	228	170	89
	PrecMax	3436	230	248	312	428	490	618	639	739	704	577	449	186
	PrecMin	1611	23	40	69	87	68	145	102	59	74	53	28	7
	PrecStd	479	51	59	61	84	111	123	127	174	167	113	100	46
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	16.0	17.6	18.9	21.9	24.4	27.1	30.1	31.1	29.8	26.1	22.4	18.7
		MCWB	14.1	15.9	15.9	18.0	19.7	24.6	26.3	26.6	25.9	23.2	19.7	16.8
	2%	DB	14.4	15.8	17.7	20.6	23.3	25.7	29.2	30.1	28.6	24.9	21.0	16.9
		MCWB	12.2	14.1	15.2	17.3	19.5	23.6	25.9	26.3	25.3	22.3	18.1	14.2
	5%	DB	12.9	14.3	16.6	19.7	22.5	24.9	28.5	29.4	27.6	24.0	20.0	15.6
		MCWB	10.0	12.4	14.1	16.7	19.1	23.1	25.6	26.0	24.5	21.2	17.3	12.6
	10%	DB	11.6	12.9	15.6	18.8	21.7	24.2	27.8	28.7	26.8	23.1	19.1	14.4
		MCWB	8.5	10.2	13.0	15.8	18.7	22.7	25.2	25.7	23.9	20.0	16.4	11.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	15.1	16.6	17.7	20.2	22.5	25.2	27.2	27.7	26.9	24.8	21.1	17.6
		MCDB	15.8	17.1	18.0	20.6	23.0	26.1	29.0	29.6	28.4	25.3	21.7	18.3
	2%	WB	13.0	14.6	16.2	18.9	21.4	24.5	26.6	27.1	26.1	23.4	19.3	15.2
		MCDB	14.0	15.6	17.0	19.8	22.1	25.2	28.2	29.0	27.6	24.3	20.1	16.2
	5%	WB	10.7	12.9	14.8	17.7	20.6	23.9	26.1	26.6	25.5	22.1	18.1	13.3
		MCDB	12.2	14.1	16.2	18.8	21.5	24.5	27.7	28.4	26.8	23.3	19.5	15.0
	10%	WB	8.9	10.8	13.5	16.7	19.7	23.2	25.6	26.2	24.9	20.9	17.0	11.8
		MCDB	11.2	12.4	15.2	18.2	21.0	24.0	27.1	27.9	26.2	22.5	18.8	14.1
Mean Daily Temperature Range	5% DB	MDBR	4.9	5.3	5.6	5.0	4.5	3.7	3.7	4.0	3.9	4.0	4.3	4.7
		MCDBR	5.7	5.8	5.8	5.3	5.1	4.3	4.5	4.6	4.5	4.3	4.4	5.1
		MCWBR	6.2	6.2	6.0	4.9	4.3	3.1	3.0	3.0	3.2	3.9	4.7	5.4
	5% WB	MCDBR	5.5	5.8	5.6	4.7	4.3	3.5	4.2	4.3	4.3	4.1	4.2	5.0
		MCWBR	6.7	6.7	6.6	5.0	4.2	2.9	2.9	2.9	3.2	4.1	5.0	6.1
Clear-Sky Solar Irradiance	taub	0.379	0.424	0.478	0.503	0.514	0.527	0.521	0.495	0.469	0.442	0.404	0.373	
	taud	2.258	2.144	2.001	1.988	2.039	2.087	2.151	2.224	2.266	2.250	2.273	2.295	
	Ebn at Noon	815	813	797	796	791	777	779	794	799	787	782	799	
	Edh at Noon	109	136	169	178	171	163	152	139	128	121	107	100	
All-Sky Solar Radiation	RadAvg	2.63	3.21	4.18	4.87	5.11	4.35	5.22	5.39	4.23	3.41	2.69	2.38	
	RadStd	0.21	0.26	0.30	0.53	0.54	0.54	0.57	0.44	0.37	0.37	0.25	0.16	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (19 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page